

MODULE 1

Answer ANY TWO questions.

1. The graph in Figure 1 shows how the population of a short-lived species varies with time. Study the graph carefully and answer the questions that follow.

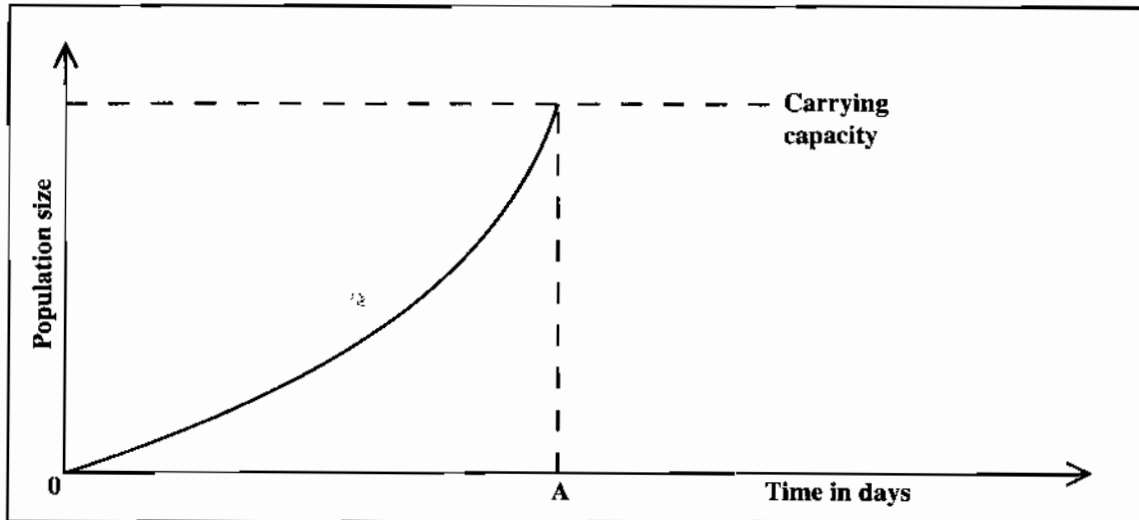


Figure 1

- (a) Identify the type of growth exhibited by this population of short-lived species, during the time period OA. [1 mark]
- (b) (i) What is meant by the term 'density dependent factors'?
- (ii) Describe the impact of TWO density dependent factors on the growth rate of the population during the time period OA. [8 marks]
- (c) Sketch and complete the graph to show how you would expect the population to grow beyond time at A. [3 marks]
- (d) Explain the shape of your graph beyond time at A, making clear in your explanation the meaning of the term 'carrying capacity'. [8 marks]

Total 20 marks

2. (a) Draw a labelled diagram to show how nutrients in the soil are recycled between the various organisms in an ecosystem. **[6 marks]**
- (b) Use your diagram to explain why natural ecosystems are self-sustaining. In your explanation use specific examples of the organisms involved in EACH stage of the process. **[6 marks]**
- (c) Explain how species diversity affects the stability of ecosystems. **[8 marks]**

Total 20 marks

3. Figure 2 shows the relationship between the size of a predator population and the size of a prey population in an ecosystem.

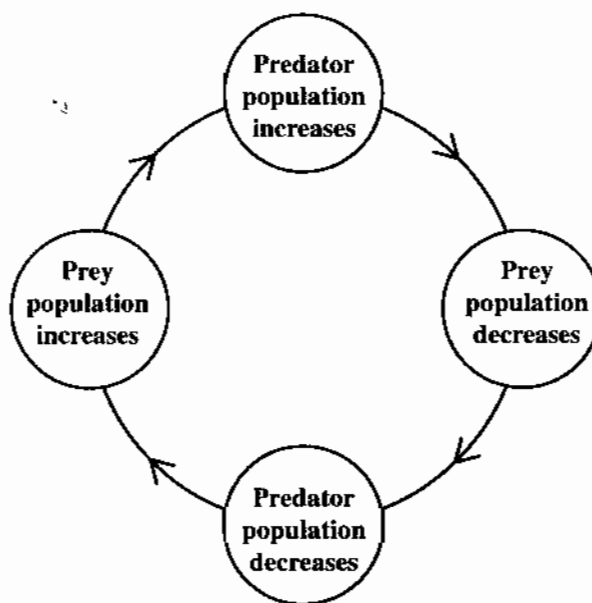


Figure 2

- (a) Explain the variations in the predator and prey populations as shown in Figure 2. **[6 marks]**
- (b) Discuss how predator-prey interactions impact on the productivity of an ecosystem. **[8 marks]**
- (c) What role do predator-prey relationships play in natural selection? **[6 marks]**

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. Figure 3 shows the growth in the world human population for both developed and developing countries for the period 1900 to 2000.

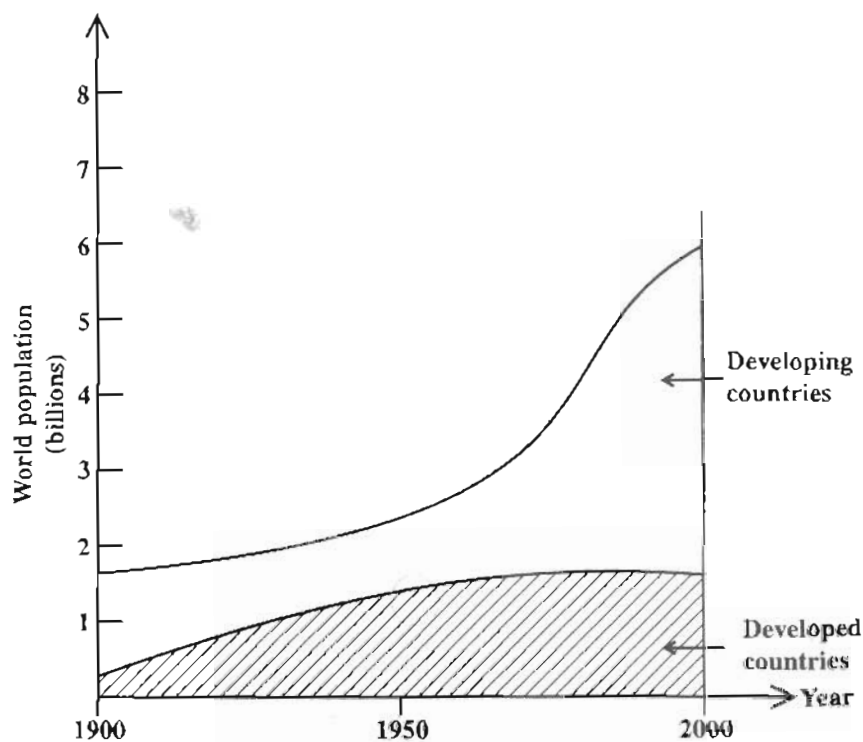


Figure 3

- (a) From the graph, describe the trend in human population growth between 1900 and 2000. [3 marks]
- (b) State TWO reasons for the rate of growth of human population since 1900. [2 marks]
- (c) Discuss TWO reasons that account for the difference in the population growth rate of developed and developing countries. [8 marks]
- (d) Assess the role of family planning in controlling the population growth in your country. [7 marks]

Total 20 marks

5. (a) Explain how EACH of the indicators below is used to quantify consumerism.
- (i) Per capita waste production
 - (ii) Per capita fuel consumption [6 marks]
- (b) Discuss how consumerism promotes EACH of the negative environmental impacts below.
- (i) Overexploitation
 - (ii) Habitat destruction [8 marks]
- (c) Assess how consumerism impacts on the ability of a named Caribbean country to achieve sustainable development. [6 marks]

Total 20 marks

6. Discuss the importance of EACH of the following as a prerequisite for sustainable development:
- (a) Population Management [7 marks]
 - (b) Environmental Management [6 marks]
 - (c) Conservation [7 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. Assess how EACH of the following factors affects the use of a named natural resource in the Caribbean:

- (a) Location [7 marks]
- (b) Role of foreign investment [6 marks]
- (c) Government policies [7 marks]

Total 20 marks

8. (a) Beaches and forests are two natural resources used in the tourism industry in the Caribbean. For EACH of these two resources, discuss TWO ways in which tourism has impacted NEGATIVELY on both beaches and forests in a NAMED Caribbean country.

[12 marks]

- (b) Discuss why the concept of ecotourism is important to the Caribbean. [8 marks]

Total 20 marks

9. (a) Explain how EACH of the measures below can be used to conserve natural resources.

(i) Environmental Impact Assessment

(ii) Protected Areas

(iii) Environmental Legislation [15 marks]

- (b) Which ONE of the measures in (a) above would be MOST effective in your country? Give ONE reason for your answer. [5 marks]

Total 20 marks

END OF TEST